OGA · Oriented Geometric Alignment & Adaptive Loss

Gaussian BEV · one independent criterion per task group · optional IoU quality supervision

Dual-stream supervision: parameter losses + joint geometry

Prediction + ground truth · heatmap, metric offset, log-size, doubled-yaw vector [+ IoU logit]

Heatmap classification
Modified focal loss
all grid cells

Offset regression
Masked Smooth-L1
positive cells

Log-size regression
Masked Smooth-L1
positive cells

Doubled-yaw regression
Masked Smooth-L1
positive cells

Oriented geometry
Lgeo = Lproj + βcorner D
derived from regression

T-SBUW · EMA calibration + learned temperature-softmax task weights
Tasks: cls, offset, size, yaw, geo [+ iou] · M=5 or 6

Repeat independently per group: separate learned task logits and EMA buffers

Optional IQA · enabled here

Quality target q = stop-gradient MGIoU
BCEWithLogits(iou_logit, q)
reg_mask only · iou_loss_weight=1

Geometry supervision · positive cells only

Decode predicted and GT corners
size = exp(clamp(log-size, −10, 10))
angle = 0.5 atan2(sin2, cos2)
center = metric offset

Four projection axes · pred + GT
Lproj = mean[(1 − meanaxis GIoU)/2]
Corner distances: direct d0, reversed d180
Normalize by GT diagonal

Dpi = mean[min(d0, d180) / diagGT] · βcorner=1 · no extra geometry head

Grouped objective

Car
Lcar

Ped / Cyc
Lvru

Ltotal = 0.5 Lcar + 0.5 Lvru

Masked Smooth-L1 divides by selected scalar coordinates + 1e−4; geometry averages positive boxes.

Empty targets: regression, geometry and IQA terms return differentiable zero; background focal supervision remains.

Q-OGA · Quality-Aligned & Range-Adaptive OGA

Gaussian BEV alternative · classification quality from decoded boxes · five adaptive tasks, without a separate IQA head

Quality-coupled heatmap + range-weighted regression + smooth geometry

Prediction + ground truth · heatmap, metric offset, log-size, doubled-yaw vector

Detached quality map q · no gradient into regression
Default mgiou: clamp(mean 4-axis GIoU, 0, 1)
Alternative rotated_iou: exact aligned polygon IoU
Optional rotated_iou curriculum: a=clip(epoch/Ewarm, 0, 1)
qpeak=(1−a)+a IoU; default Ewarm=0

Range-adaptive weights from GT
world = cell-center grid + GT offset
r = Euclidean distance to origin
wr = 1 + 1.5 clip(r/70.4, 0, 1.5)^2

Heatmap classification
CQFL(p, Y, q)
all grid cells

Offset regression
Range-weighted Smooth-L1
positive cells

Log-size regression
Range-weighted Smooth-L1
positive cells

Doubled-yaw regression
Range-weighted Smooth-L1
positive cells

Oriented geometry
Lgeo = Lproj + βcorner D
derived from regression

T-SBUW · EMA-calibrated five-task objective
Lcls + Loffset + Lsize + Lyaw + Lgeo with learned task weights

CQFL peak target = q^βq; Gaussian neighbors keep negative targets. βq=1, focal power=2.

Geometry supervision · positive cells only

Decode predicted and GT corners
size = exp(clamp(log-size, −10, 10))
angle = 0.5 atan2(sin2, cos2)
center = metric offset

Four projection axes · pred + GT
Lproj = mean[(1 − meanaxis GIoU)/2]
Corner distances: direct d0, reversed d180
Normalize by GT diagonal

Dsoft = mean[−tau logsumexp(−d0/tau, −d180/tau) / diagGT]

tau=0.05 · βcorner=1 · no extra geometry head

Grouped objective

Car
Lcar

Ped / Cyc
Lvru

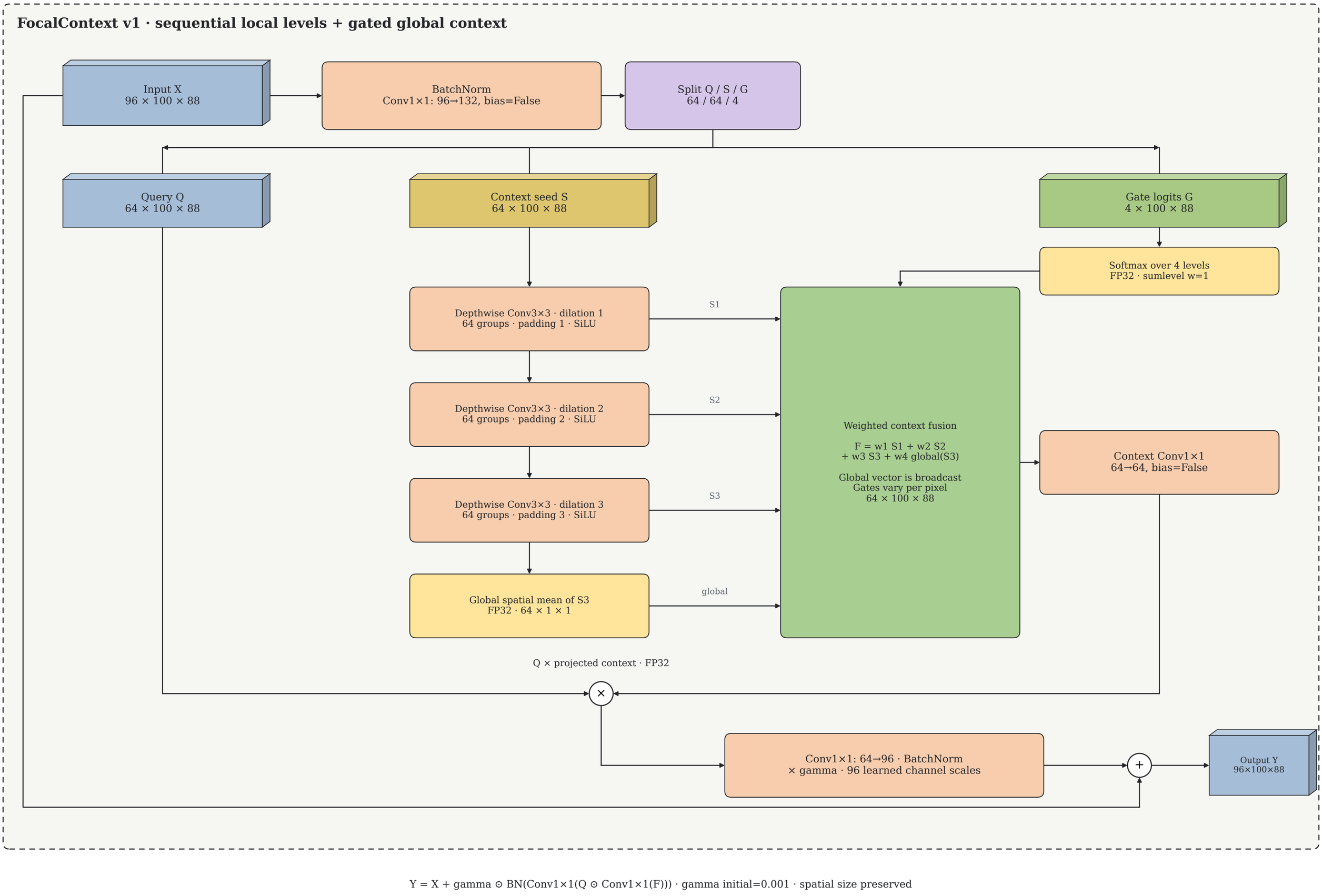
Ltotal = 0.5 Lcar + 0.5 Lvru

Coordinate loss = meanpositive[wr × sumcoordinate Smooth-L1] · range weighting applies to offset, size and yaw.

Default mgiou quality is a projection proxy; rotated_iou is area IoU. Curriculum epoch starts at zero.

Focal Context · Feature-based Context Modulation

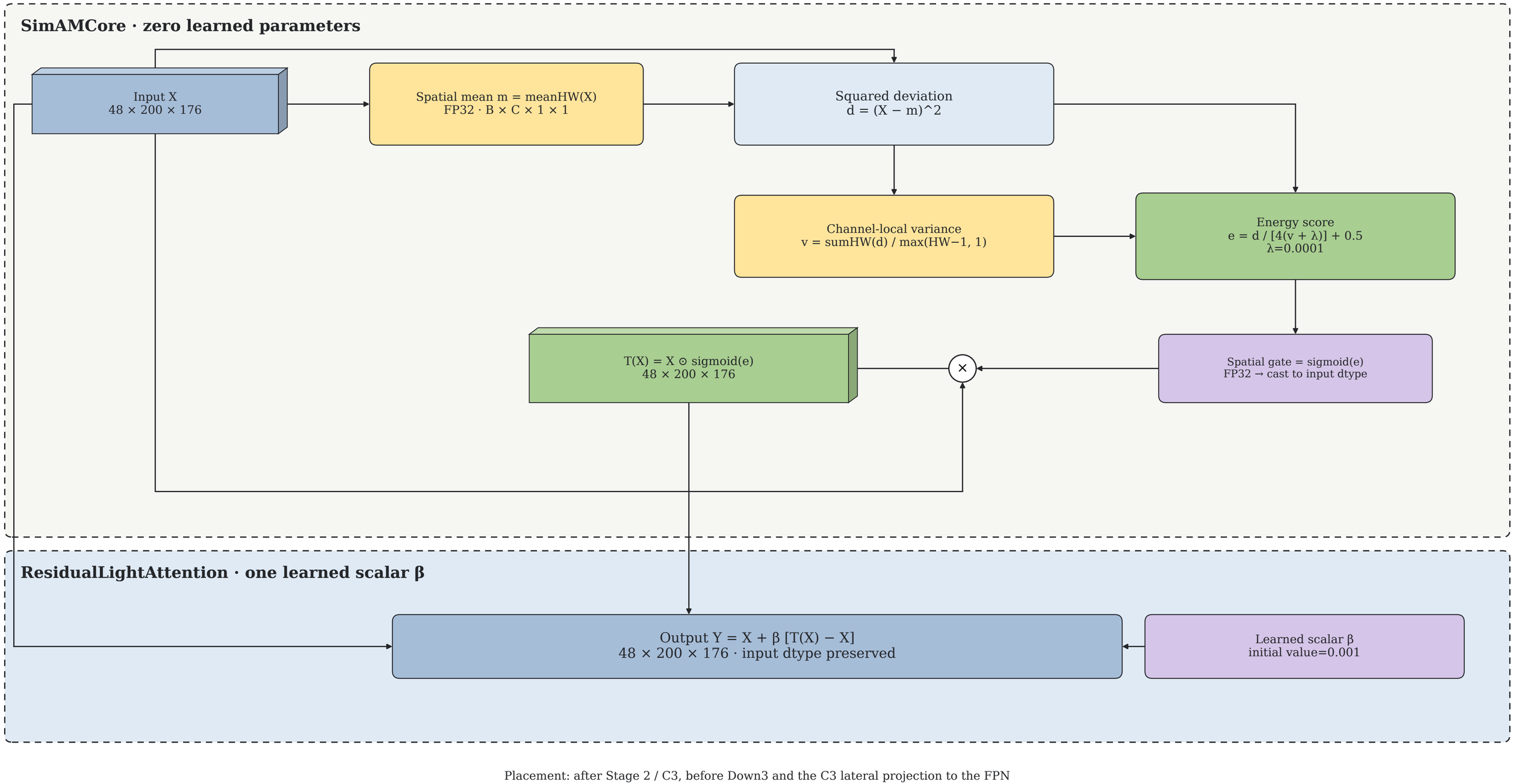
C4 / stride 8 · C=96, bottleneck D=64, dilations=(1,2,3) · 25,120 parameters



Y = X + gamma ⊙ BN(Conv1×1(Q ⊙ Conv1×1(F))) · gamma initial=0.001 · spatial size preserved

SimAM · Spatial Energy Attention + Residual Adapter

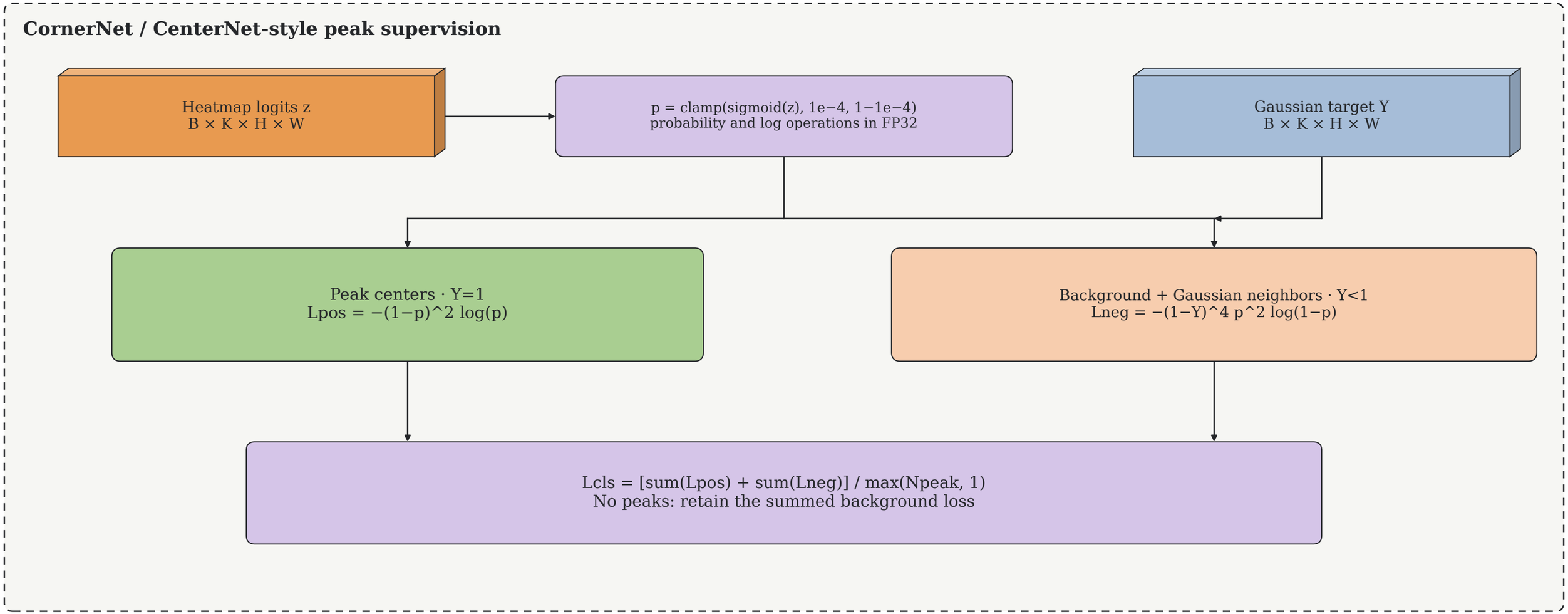
Stage 2 / C3 · 48 × 200 × 176 · per-batch, per-channel spatial statistics



Placement: after Stage 2 / C3, before Down3 and the C3 lateral projection to the FPN

Gaussian Heatmap · Modified Focal Loss

Classification term in OGA · cls_encoding=gaussian · separate from the backbone Focal Context block



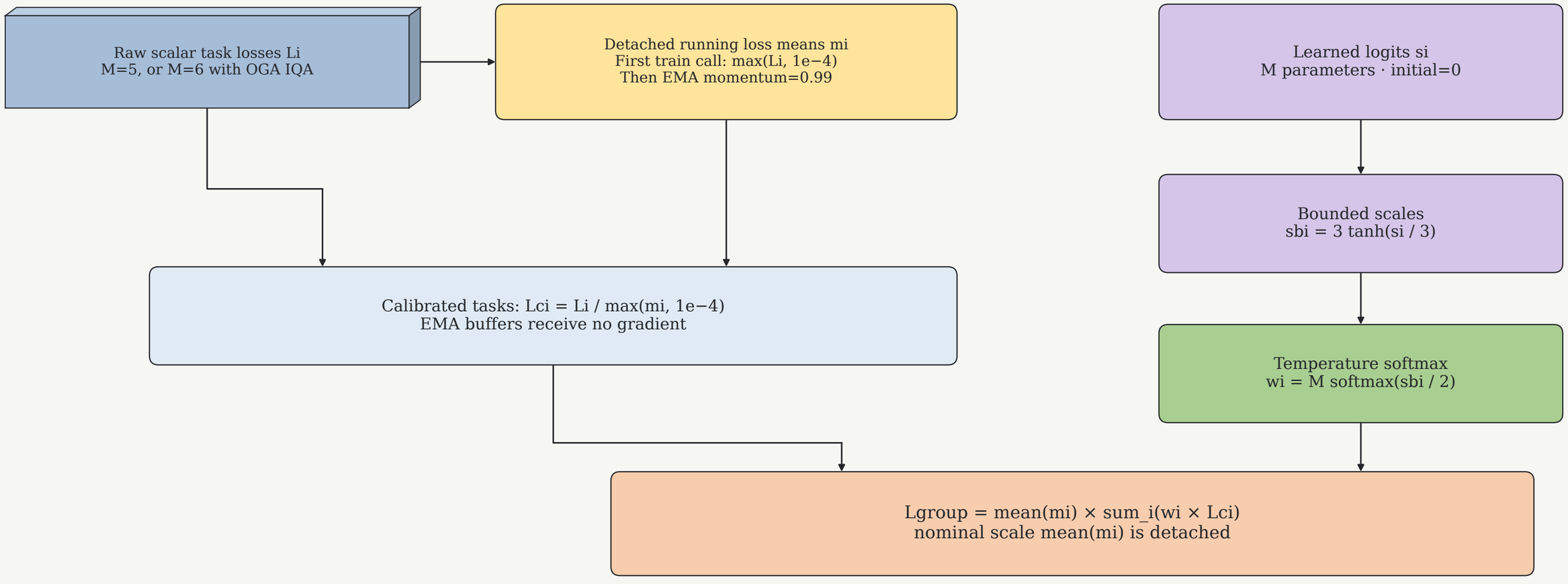
Q-OGA replaces only the peak term with |q^βq−p|^2 BCE(p, q^βq); the negative term remains the same.

At q=1, the implemented quality-coupled focal loss equals this modified focal loss.

T-SBUW · Adaptive Task Weighting with EMA Calibration

Used by both OGA and Q-OGA · learned loss weights are training-only parameters

One separate weighting module per group



All wi > 0 · sum_i wi = M · eval mode freezes EMA updates · each group owns its logits and buffers

Defaults: temperature=2 · bound=3 · EMA momentum=0.99 · loss calibration floor=1e−4

Task weights act on loss terms; they are distinct from fixed outer group weights (0.5 / 0.5 here).